

Katherine M. Braught

 <https://katherinebraught.github.io/>

Education

University of Illinois Urbana–Champaign

Ph.D. in Computer Science, Advisor: Jeff Erickson and Colleen Lewis

Area: Computers and Education

Aug 2021–present

University of Illinois Urbana–Champaign

M.S. in Computer Science, Advisor: Sayan Mitra

Area: Formal Methods for Verification

Aug 2021–Aug 2024

Iowa State University, Ames, Iowa

B.S. in Computer Science, University Honors

Summa Cum Laude, Top 2% of College

Minor: Biology

Aug 2016–Dec 2020

Research Interests

My research focuses on making students' experiences in upper-level computer science theory courses engaging and supportive. I approach this by (1) investigating the culture and experiences of students in algorithms courses, (2) developing personalized problem sets for these courses, and (3) training TAs to teach problem-solving effectively.

Teaching

Instructor Experience

University of Illinois

CS 173 Discrete Structures Instructor

Summer 2026

- Designed and delivered four 30-minute lectures every week for 75 students
- Guided students through four 45-minute group problem solving sessions every week
- Guided one TA through grading assignments, and managed in-class help from three undergraduate assistants

University of Illinois Extension Office

Principal's Scholars Program Pre-Algebra Instructor

November 2025

Developed curriculum to support pre-algebra skills for 26 6-8th grade students

Illinois Mathematics and Science Academy

Motorola CS+X Summer Campus Instructor

June 2025

Taught a 40-hour course to 20 middle school students with a CS + linguistics project. Developed instructional materials that linked Python and core computing concepts to real-world contexts

Teaching Assistant Experience

CS 374 Intro to Algs and Models of Comp, UIUC

Teaching Assistant and Discussion Section Leader

Aug 2025–May 2026

- Lead group problem-solving session with 100 students twice a week
- Support undergraduate employees' grading and grade exams

CS 591 Teaching Assistant Training, UIUC

Course Assistant

Aug 2024–present

- Manage attendance of 300 students
- Grade reflection assignments
- Contribute to improving TA training lectures

CS 173 Discrete Structures, UIUC

Teaching Assistant

Aug 2024–May 2025

- Teach weekly tutorials (200+ students) and hold office hours
- Grade biweekly proof-based exams

Guest Lecturing.....

Teaching and Leadership Eng 590

Teaching Engineering Problem Solving and Meeting Students Where They Are

Sp 26, Fa 25 & Sp 24

Lectured approx. 300 graduate teaching assistants on how to teach problem solving

UIUC's Grad Academy

Using Compassionate Pedagogy as a TA

Fall 2025

Using active learning, workshopped with about 60 students on using compassionate pedagogy

UIUC's Grad Academy

Teaching Computer Science Problem Solving and Meeting Students Where They Are

Spring 2025

Lectured approx. 40 computer science graduate teaching assistants on how to teach problem solving

Professional Work Experiences

IBM

Backend Developer: Systems

Jan 2021–Aug 2021

- Developed software in proprietary language for IBMi
- Implemented new data type support for databases

Research Internship Experience

NASA Langley

PhD Student Researcher

Mentor: Alwyn Goodloe, Ph.D.

May 2023–Aug 2023

- Implemented probabilistic monitoring in Copilot
- Developed monitoring examples for safety-critical systems

Air Force Research Lab, Wright-Patterson AFB

PhD Student Researcher

Mentor: Kerrianne Hobbs, Ph.D.

May 2022–Aug 2022

- Synthesized a reinforcement learning controller for the Inspection Problem
- Integrated runtime assurance into RL controllers

Peer-Reviewed Work

Student Experiences in Algorithms courses

SIGCSE

Katherine Braught, Colleen Lewis, Jeff Erickson

2027

Under review

Novice Difficulties with Graph Problem Reduction in Algorithm Design

SIGCSE virtual

Hongxuan Chen, Katherine Braught, Geoffrey Herman and Jeff Erickson

2026

Accepted

Evaluating LLM-Generated Contextualized Algorithm Design Problems

ICER

Erica Goodwin, Katherine Braught* et al. , *equal contribution*

2026

DOI: 10.1145/3765964.3811658

An Engineering TA training Lesson on Problem Solving: Design and Evaluation

ASEE

Katherine Braught, Blake Johnson, Yael Gertner

2026

Accepted

Interdisciplinary Engineering Education in a Middle School Computer Science Camp

ASEE

Kathleen Isenegger, Katherine Braught et al.

2026

Accepted

A TA Training Lesson for Problem-Solving

SIGCSE

Katherine Braught, Carl Evans, Blake Johnson, Yael Gertner

2026

DOI: 10.1145/3770761.3777214

Too Theoretical and Nowhere Near Interesting (Formal Methods)

ITICSE

Katherine Braught, Yangge Li, Katherine Driggs-Campbell, Sayan Mitra

2025

Using a Tool to Increase Student Motivation for Formal Methods

DOI: 10.1145/3724363.3729116

Novice Difficulties in Graph Layering for Algorithm Design <i>Hongxuan Chen, Katherine Braught, Geoffrey L. Herman, Jeff Erickson</i> DOI: 10.1145/3641555.3705221	SIGCSE 2025
Creating a Community of Graduate Student Computer Science Education Researchers <i>Grace Barkhuuff, Katherine Braught, Emma R. Dodoo, Michael Link, Xinying Hou, Elliot Roe</i> DOI: 10.1145/3641555.3705111	SIGCSE 2025
Verse: A Python Library for Reasoning About Multi-agent Hybrid System Scenarios <i>Y. Li, H. Zhu, K. Braught, K. Shen, S. Mitra</i> DOI: 10.1007/978-3-031-37706-8_18	CAV 2023
Structural and Sequential Cause in Hidden Markov Models <i>K. Braught, S. Mitra</i> Accepted submission only	CREST 2023
Checking Phylogenetics Decisiveness in Theory and in Practice <i>Ghazaleh Parvini, Katherine Braught, David Fernández-Baca</i> DOI: 10.1109/TCBB.2021.3128381	TCBB 2022

Honors and Awards

2026: Graduate Teacher Certificate, in progress
2025-2026: University of Illinois Mavis Future Faculty Fellow
2025: UIUC Academy for Excellence in Engineering Education Recognition of Outstanding Student Contribution to a Strategic Instructional Innovations Program project

Service, Leadership, and Outreach

Computer Science Culture and Climate Committee <i>UIUC</i>	2026
Reviewer <i>ITICSE</i>	2025, 2026
Reviewer <i>SIGCSE</i>	2026
Scholarship Reviewer <i>UIUC Outreach Workshops</i> Work with local students to review scholarship applications to NCWIT Aspirations in Computing	2024-Present
Volunteer Instructor <i>UIUC Summer Camps</i> Worked with Illinois students for a two day workshop at a summer engineering camp	Summer 2024
Lead Volunteer Instructor <i>UIUC CS+X Workshops</i> Work with local middle and high school students to teach CS+X workshops	2023-Present

Personal Service

Election Judge <i>Champaign County Clerk Office</i> ensured fair, free, and accessible election while administering the election process in Champaign county	2024-present
Member <i>Junior League of Champaign-Urbana</i> Partnered with local non-profits and coordinates service projects	2023-2025
Special Events Volunteer <i>Champaign Parks Department</i> Supported large community events and resident engagement	2022-2024